EXASOL EXATHON · TRACK B · MONEY, RISK & FINANCIAL DERIVATIVES

LiqShield

Will we survive Friday's margin calls?

Collateral & liquidity stress intelligence: eligible collateral, shortfall timing and funding cost, computed in SQL.

Wolves of Bharathi Salai · October 2026

Headline collateral overstates what you can actually pay with

Encumbered

Assets already pledged elsewhere cannot be posted again. In our book, $8M of AAA bonds is locked.

Haircuts

Collateral is worth less than market value: 0% on cash, 2% on Treasuries, 25% on equities.

Settlement lag

Bonds take a day to become cash, equities two. A call on T+1 cannot use T+2 money.

Late inflows

Incoming settlements slip while margin calls grow. Stress hits all of it at once.

$32.0M → $22.55M on the books versus what can actually be mobilised

One question, four numbers

Can we meet time-sensitive obligations with the cash and collateral we can actually mobilise?

$22.55M

Eligible collateral after haircuts and encumbrances

-$4.49M

Shortfall severity: the deepest buffer breach under severe stress

T+4

Time to shortfall: the first day the buffer goes negative

≈ $1.2k

Intervention cost: the cheapest in-time funding that closes the gap

Four SQL steps, no application maths

1

Eligibility

Join collateral to haircut rules, drop encumbered assets, apply the stress haircut.

2

Time ladder

Bucket cashflows T+0 to T+7, shift inflows, scale margin calls, window-sum the buffer.

3

Shortfall metrics

Severity, first breach day and days below zero in a single pass.

4

Funding waterfall

Rank facilities by cost, exclude those too slow, draw until the gap is closed.

The core of the ladder, one window function:

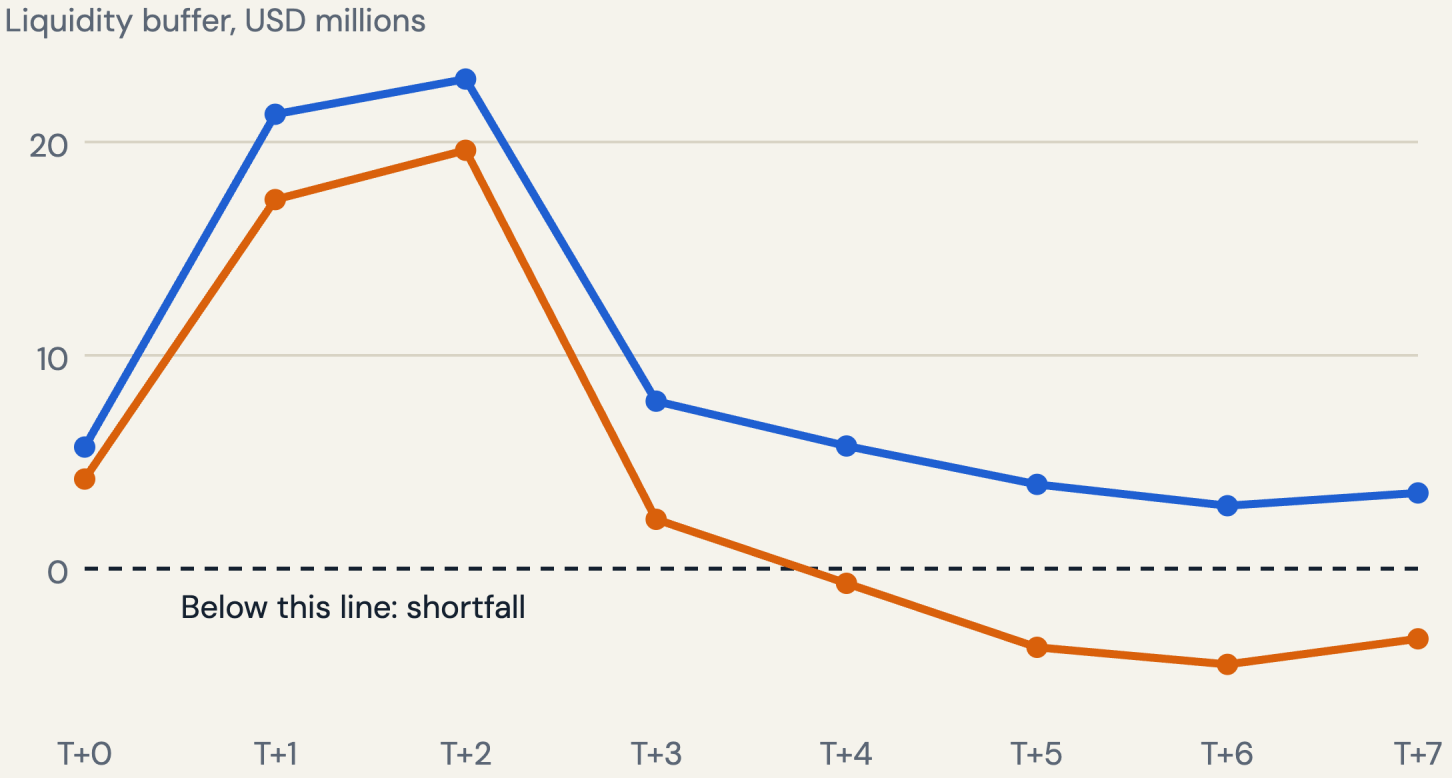
```
MOBILISABLE + SUM(NET_CASHFLOW) OVER (ORDER BY DAY_OFFSET) AS LIQUIDITY_BUFFER
```

Same book, three scenarios

Scenario	Haircut	Delay	Margin calls	Lowest buffer	First breach	Peak gap
Base	+0 pp	0 days	×1.00	$2.95M	none	none
Moderate	+5 pp	1 day	×1.10	−$0.27M	T+5	$0.27M
Severe	+10 pp	2 days	×1.20	−$4.49M	T+4	$4.49M

Synthetic data, as of 2 Nov 2026. Mobilisable collateral is $22.55M. The designed stress event is a $15M Bank_B margin call on T+3: survivable in the base case, a breach once haircuts, delays and margin scaling stack up.

Buffer by day: base vs severe stress



Base case

Severe stress

Severe stress first goes below zero on T+4 and bottoms out at −$4.49M on T+6.

The T+3 margin call is survivable in the base case. Add +10 pp haircuts, late inflows and calls ×1.2, and it is not.

Funding actions: cheapest first, only if in time

Facility	Limit	Available	Annual cost	Severe stress (breach T+4)
Central-bank credit line	$5M	T+0	2.50%	Draws $4.49M, cost ≈ $1.2k
Committed revolver	$8M	T+1	3.20%	Available, not needed
Overnight repo line	$10M	T+1	4.50%	Available, not needed
Equity fire-sale	$3M	T+2	12.00%	Available, not needed
Parent support (uncommitted)	$20M	T+4	6.00%	Available, not needed

Lead time matters: in a tougher case (+20 pp haircuts, 2-day delay, calls ×1.3) the breach moves to T+3, and parent support, available only from T+4, is excluded as too late.

Every number is checked three ways

1

Hand calculation

```
5.00 + 9.80 + 3.80 + 1.70
+ 2.25 = $22.55M eligible
```

Cash, Treasuries, AAA, BBB and equities after haircuts, with the pledged $8M excluded.

2

Independent oracle

A pure-Python re-implementation shares no code with the SQL.

11 pytest tests compare the two across four scenarios, plus hand-calculated constants.

3

Excel model

A formula-driven workbook recomputes the ladder from first principles.

Switch scenario and the PASS cell confirms a match to the cent, all 8 buckets.

Built for Exasol, honest about status

Designed to run in the database

- Joins, filters and window-function ladders in SQL
- Funding waterfall using running window sums
- Python UDF for Monte Carlo haircut shocks, inside the database
- pyexasol connector, schema, loader script included

Where we are today

- Verified end to end on DuckDB, running the identical SQL
- 11 of 11 tests pass; Excel shows PASS in all three scenarios
- Our local Exasol Personal deployment would not start on submission night
- The Exasol path is implemented but not yet verified on a live instance

Next: run it on Exasol at scale

- Verify the Exasol path and the in-database UDF on a live Exasol Personal deployment
- Scale to millions of positions and intraday time buckets
- Add multi-currency, netting sets and counterparty limits
- Connect live collateral and cashflow feeds

Code: github.com/krithiyaragarla/liq_shield_exasol

Synthetic data (CC0) · MIT-licensed code